

Modeling Kinematic Signatures of Aging via Diffusion and Graph Networks

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Abstract

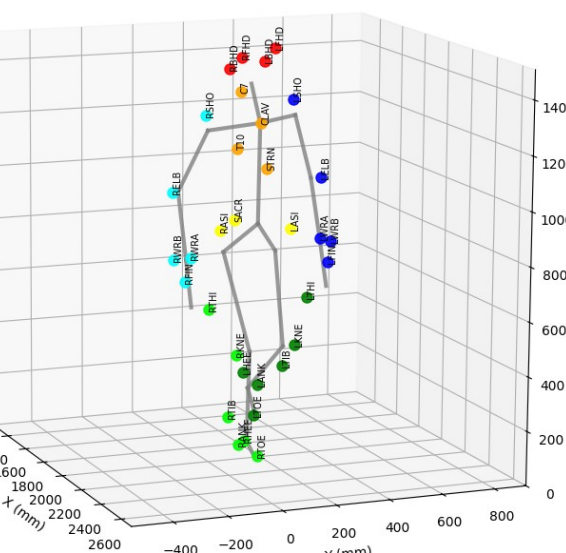
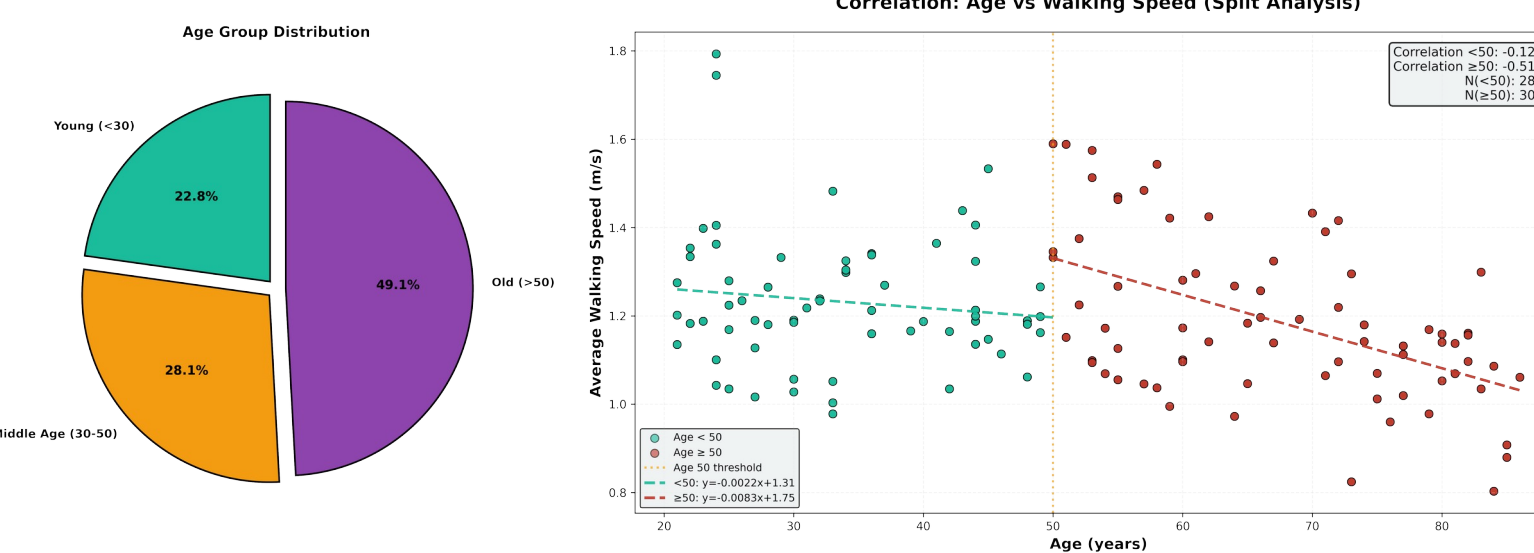
It is well established that aging significantly shapes human movement. Yet existing text-to-motion generative models lack the resolution to reproduce these subtle but clinically meaningful differences. We ask: *can data-driven models learn age-related biomechanical variation, and generate motions that faithfully reflect those differences?*

We present a three-component framework:

- A pipeline transforming clinical C3D motion-capture recordings into the machine-learnable graph representations (*NTU3D* and *HumanML3D*), while preserving age-related kinematic cues.
- A **Spatio-Temporal Graph Convolutional Network (ST-GCN)** detects aging signature and extracts a latent age embedding **z-age** from skeletal motion sequences.
- This embedding **z-age** is injected into a text-conditioned **Motion Diffusion Model (MDM)** via cross-attention, enabling age-conditioned motion generation from natural-language prompt.

Dataset: Van Criekeing

58 Data Points 138 Subjects 21–86 Age Range 100Hz Capture 588 Trials

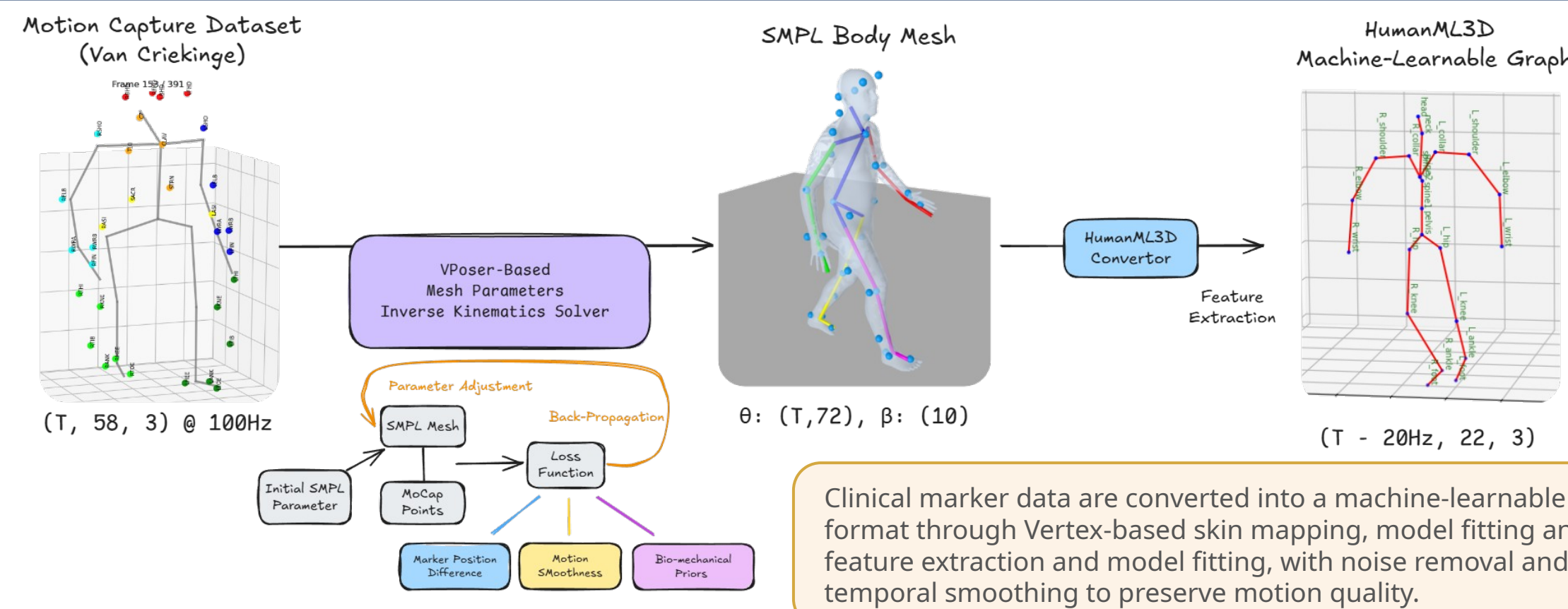


Age vs. Walking Speed Correlation

Under 50: $r = -0.13$ (negligible correlation)
Over 50: $r = -0.51$ (pronounced decline emerges)

Reduced is one of the primary kinematic signatures of age. While plain walking speed accounts for 50% correlation in aging, **discrete latent embeddings** from ST-GCN capture structural gait differences — posture, coordination, and joint dynamics.

Data Pipeline



Model Architecture

A **Spatio-Temporal Graph Convolutional Network** trained to discriminate between three age groups: Young (<30), Mid-age (30–60), and Elderly (>60) from skeleton motion.

The backbone is initialized from weights pretrained on NTU RGB+D, while the modified classifier head is retrained on Van Criekeing data.

A compact classification head (256–32–3) produces age-group predictions; the 32-dimensional penultimate layer serves as **z-age** — a biomechanically-grounded latent encoding of gait age.

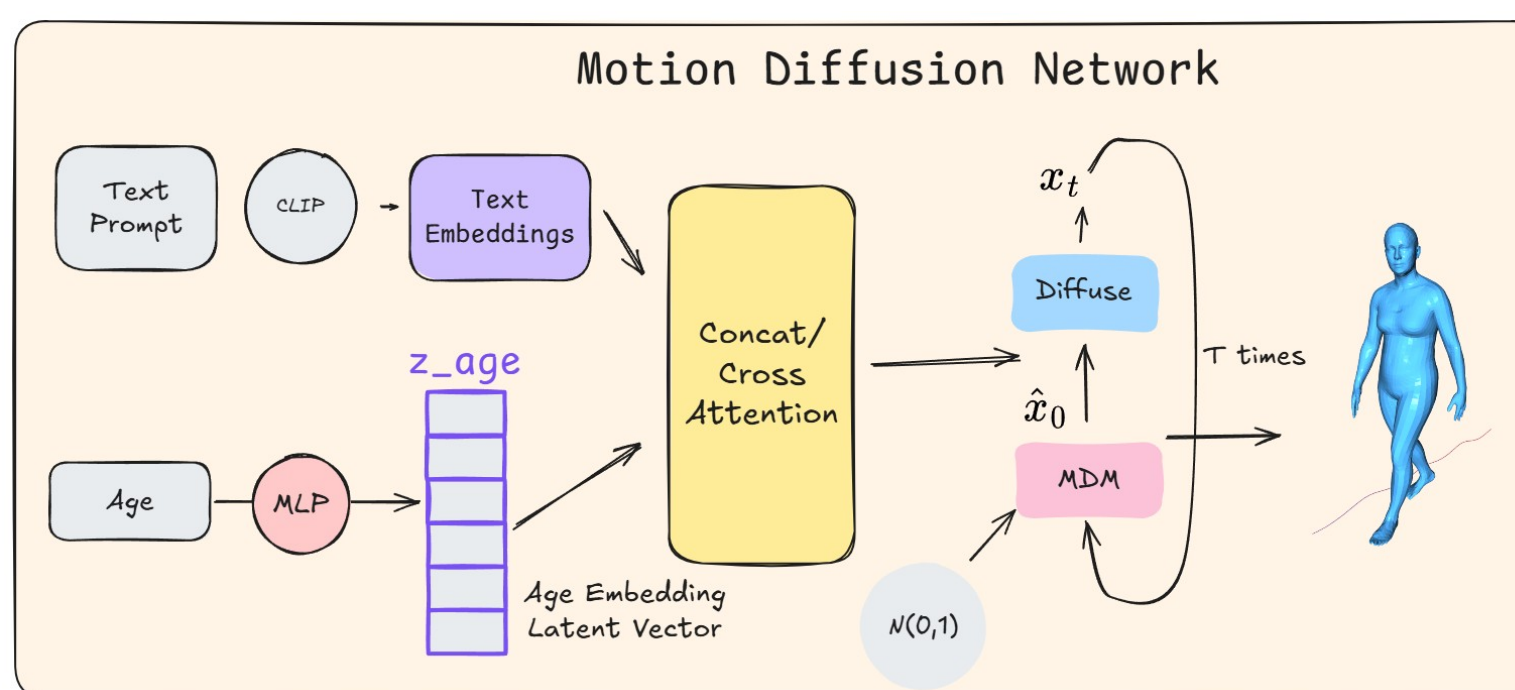
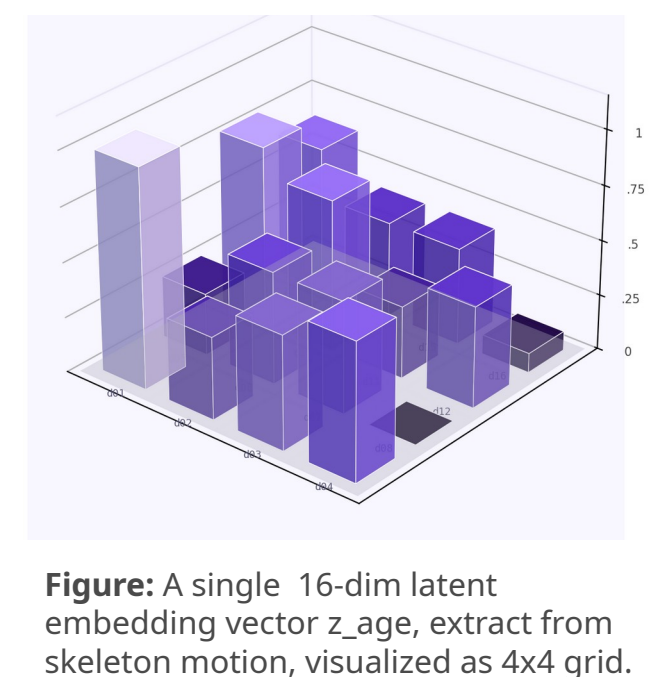
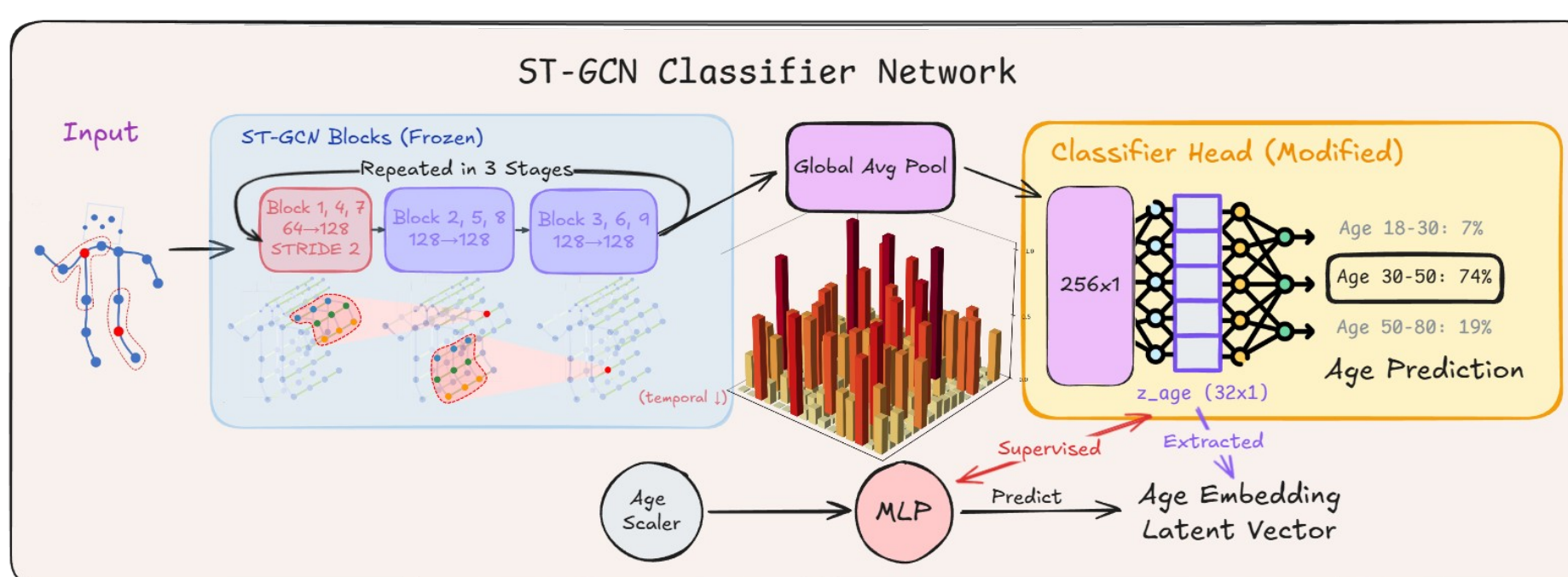
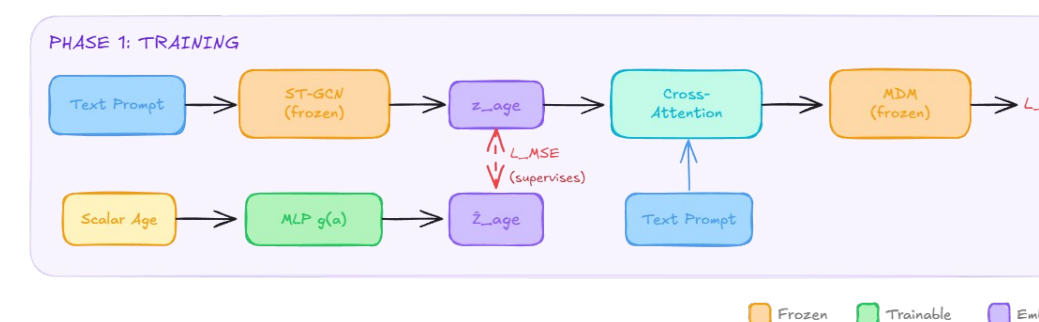


Figure: During inference, no input motion clip is required. The scalar age is mapped to z_{age} by the trained MLP alone, enabling generation from text prompt and age without real reference motion.

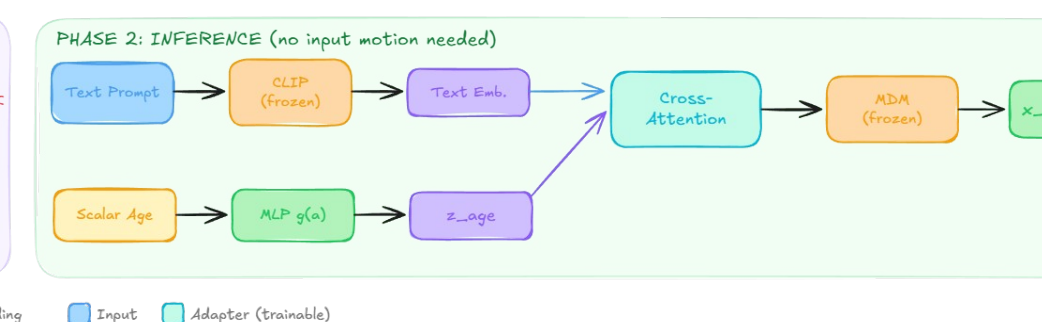


The diffusion model follows the Human Motion Diffusion Model (MDM) architecture. The forward diffusion process is:

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_0, (1 - \alpha_t)I)$$

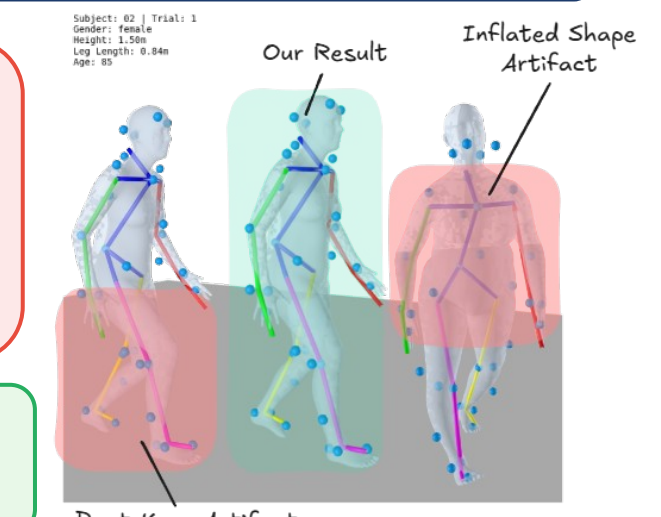
where $x_t \in \mathbb{R}^B \times T \times 263$ is the noisy motion sequence. Age conditioning is injected by concatenating z_{age} with the text embedding in the cross-attention layers of the denoising transformer. The training objective minimizes the simple noise-prediction loss:

$$\mathcal{L} = \mathbb{E}_{x_0, \epsilon, t} [\|\epsilon - \epsilon_\theta(x_t, t, z_{text}, z_{age})\|^2]$$



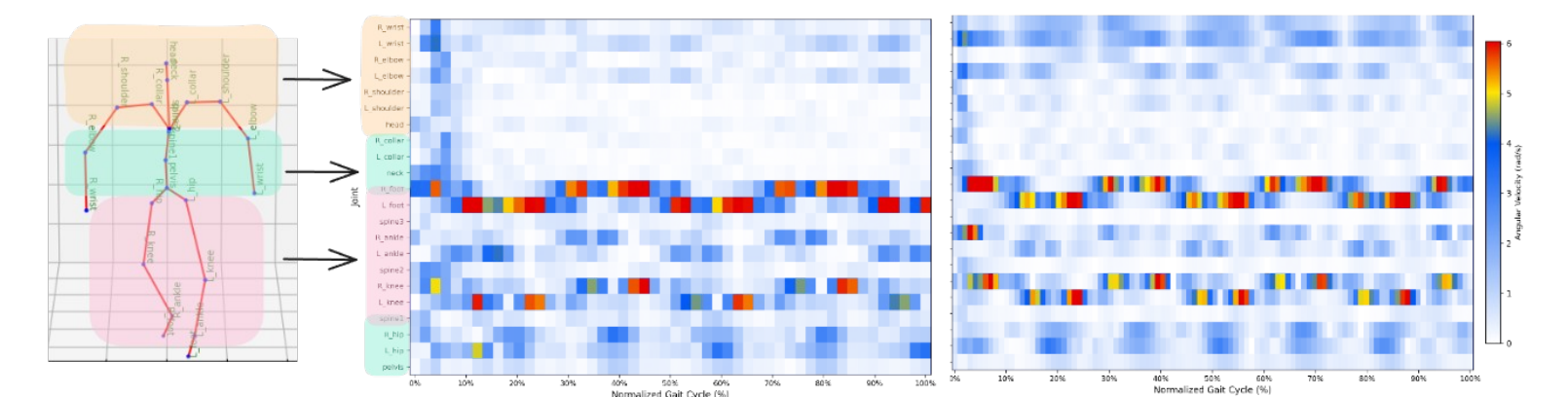
Pipeline Validation

- Mesh Inflation**
Over-estimated β params → unrealistic proportions
- Crouch Walking**
Spurious knee flexion from marker occlusion
- Our Pipeline (VPoser-based)**
Learned pose prior regularizes solution



Age-discriminative kinematic signals are preserved after conversion — confirming pipeline fidelity. Joint angular velocity (rad/s) across a normalized gait cycle. Elderly subjects (Age 78) show systematically reduced peak velocities and constrained range of motion versus young subjects (Age 24).

- Age 24 (Young)**
 - Higher peak velocities
 - Standard gait pattern
- Age 78 (Elderly)**
 - Reduced peak velocity
 - Lower amplitude across joints



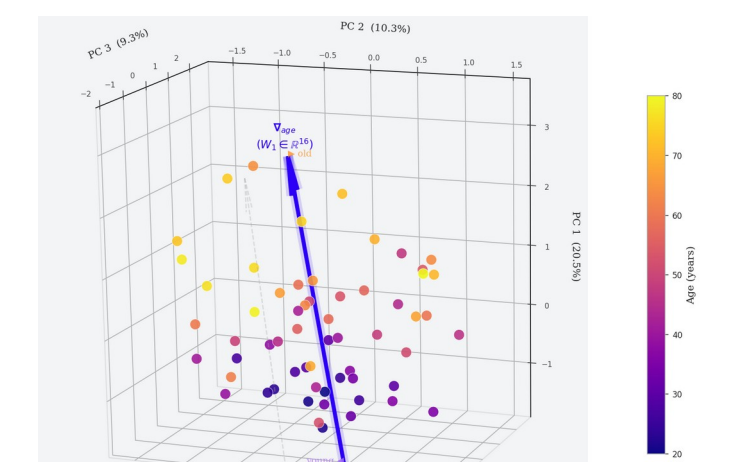
Limitations:

- Small and imbalanced dataset: 138 subjects with skewed age-group distribution.
- Training Pipeline in refinement: trained ST-GCN classifier model shows moderate-low accuracy and is being improved.

Future Directions:

- Complete the generation loop: Integrate trained z_{age} embeddings into the LoRA-MDM cross-attention pipeline
- Age-specific evaluation protocol: Develop metrics grounded in clinical gait analysis to measure biomechanical plausibility across age groups beyond FID.

Figure: PCA of z_{age} embeddings (60 subjects, ages 20–80) reveals moderate-weak age-correlated structure in the current latent space, indicating the classifier has not yet learned age-discriminative representations. This motivates refinement in model architecture and retraining with larger, more balanced and action-rich dataset.



Acknowledgments

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Key references:

- [1] Guo et al. — Generating Diverse and Natural 3D Human Motions from Text. CVPR 2022.
- [2] Tevet et al. — Human Motion Diffusion Model. ICLR 2023.
- [3] Yan et al. — Spatial Temporal Graph Convolutional Networks for Skeleton-Based Action Recognition. AAAI 2018
- [3] Van Criekeing et al. — A full-body motion capture gait dataset of 138 able-bodied adults across the life span and 50 stroke survivors.